Lecture 5: Conditional Probability & Multiplication Theorem

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$S = \{1, 2, 3, 4, 5, 6\}$

Topics of This Lecture:

1. Conditional Probability
2. Multiplication Theorem

Random Experiment: You roll a pair of 6 sided dice.

Sample Space(S) : { (1,1) (1,2) (1,3) (1,4) (1,5) (1,6)
(2,1) (2,2) (2,3) (2,4) (2,5) (2,6)
(3,1) (3,2) (3,3) (3,4) (3,5) (3,6)
(4,1) (4,2) (4,3) (4,4) (4,5) (4,6)
(5,1) (5,2) (5,3) (5,4) (5,5) (5,6)
(6,1) (6,2) (6,3) (6,4) (6,5) (6,6) }

$S = \{(1,1)\ (1,2)\ (1,3)\ (1,4)\ (1,5)\ (1,6)$
$(2,1)\ (2,2)\ (2,3)\ (2,4)\ (2,5)\ (2,6)$
$\vdots$
$(6,1)\ (6,2)\ (6,3)\ (6,4)\ (6,5)\ (6,6)\}$

$|S| = 36$

$|S| = 36$

Q1. What is the probability that sum of numbers on two dice is 6 ?

Event (E) = sum of numbers on two dice is 6
(E) = {(1,5) (2,4) (3,3) (6,2) (5,1)}
|E| = 5

P(E) = |E| / |S|
P(E) = 5 / 36

$E = \{(1,5)\ (2,4)\ (3,3)\ (6,2)\ (5,1)\}$

$|E| = 5$

$P(E) \frac{|E|}{|S|} = \frac{5}{36}$

Probability

$P(E) = \frac{|E|}{|S|}$

- E
- F

$\boxed{P(E|F)} = ?$

0

Conditional Probability

$P(E|F) = ?$

F

Q2. In above example, given that the first die lands on 2.
What is the probability that sum of numbers on two dice is 6 ?

0

Event (E) = sum of numbers on two dice is 6
E = {(1,5) (2,4) (3,3) (4,2) (5,1)}
|E| = 5

Event (F) = The first die lands on 2.
F = {(2,1) (2,2) (2,3) (2,4) (2,5) (2,6)}
|F| = 6

E ∩ F = { (2,4) }
|E ∩ F| = 1

P(E|F) = |E ∩ F| / |F| = 1 / 6

$P(E|F) = \frac{|E \cap F|}{|F|} = \frac{\frac{|E \cap F|}{|S|}}{\frac{|F|}{|S|}} = \frac{|E \cap F|}{|F|}$

$\boxed{P(E|F) = \frac{|E \cap F|}{|F|}}$

$= \frac{1}{6} = 16.66\ \%$

P(E|F) = P(E ∩ F) / P(F)

$= \frac{|E \cap F| / |S|}{|F| / |S|}$

= |E ∩ F| / |F|

6

$P(E|F) = P(E \cap F) / P(F)$

$= \dfrac{|E \cap F| / |S|}{|F| / |S|}$

$= |E \cap F| / |F|$

If E & F are two events,

Conditional Probability of event E
Given that event F has already happened
Is denoted by $P(E|F)$

$$P(E|F) = P(E \cap F) / P(F)$$

Q3. On rolling a die, what is the probability of getting a prime number given that an even number showed up.

Event (E) = Getting a prime number
E = {2, 3, 5}
|E| = 3

Event (F) = An even number showed up
F = {2, 4, 6}
|F| = 3

$E \cap F = \{2\}$
$|E \cap F| = 1$

$P(E|F) = |E \cap F| / |F| = 1/3$

P(E) = getting a prime no.

$E \cap F = 1$

$P(E|F) = \dfrac{|E \cap F|}{|F|} = \dfrac{1}{3} = 33.33\%$

Q4. Random Experiment: Rolling 2 dice together
E = Sum of the numbers is 8
F = Both die show on even no

1) $P(E|F) = ?$

2) $P(F|E) = ?$

E = {(2,6) (3,5) (4,4) (6,2) (5,3)}
|E| = 5

F = { (2,2) (2,4) (2,6)
(4,2) (4,4) (4,6)
(6,2) (6,4) (6,6) }
|F| = 9

$E \cap F = \{(2,6)\ (4,4)\ (6,2)\}$
$|E \cap F| = 3$

- E + F = 8
- E & F → even no.

$|E \cap F| = 3$
$|F| = 9$
$|E| = 5$

① $P(E|F) = \dfrac{|E \cap F|}{|F|} = \dfrac{3}{9} = \dfrac{1}{3} = 33.33\%$

② $P(F|E) = \dfrac{|E \cap F|}{|E|} = \dfrac{3}{5} = 60\%$

1) $P(E|F) = P(E \cap F) / P(F)$
$= 3/9 = 1/3$
$= 33.33\ \%$

2) $P(F|E) = P(F \cap E) / P(E)$
$= 3/5$
$= 60\ \%$

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Conditional Probability

Q5. A box contains 4 white balls 3 red balls,
Two balls are randomly selected & removed from the box 0 back to back.
Given that first removed ball is white.
What is the probability that the 2nd removed ball is red?

$A \cap B = \dfrac{4}{7} \times \dfrac{3}{6} = \dfrac{2}{7}$

$\dfrac{4}{7}$ $\dfrac{3}{6}$

$\frac{4}{7}$ $\frac{3}{6}$

Sample Space = { ○○○○●●● }
|S| = 7

Event A = 1st ball is white
P(A) = no. of white balls / no. of elements in sample space
P(A) = 4 / 7

After removing first ball (given white)
Sample Space = { ○○○●●● }

Event B = 2nd ball is red
P(B) = no. of red balls / no. of elements in sample space
P(B) = 3 / 6 = 1/2

P(B ∩ A) = (4/7) X (1/2)

P(B|A) = P(B ∩ A) / P(A)

$$= \frac{(4/7) \times (1/2)}{(4/7)}$$

= 1/2

$$P(B|A) = \frac{|A \cap B|}{|A|}$$

$$= \frac{\frac{2}{7}}{\frac{4}{7}} = \frac{2}{\cancel{7}} \times \frac{\cancel{7}}{4} = \frac{1}{2}$$

$= 50\%$

Q6. A box contains 25 Bipolar Junction Transistors(BJT),
5 of them are defective,
4 of them are partially defective (fail after few hours of uses),
16 of them are acceptable.
A BJT is chosen randomly for use, it doesn't fail immediately.
What is the probability that it is acceptable?

Number of Elements in Sample Space |S| = 25
Defective BJTs --> Event A
Partially Defective BJTs --> Event B
Acceptable BJTs --> Event C

|A| = 5
|B| = 4
|C| = 16

|S| = 25 - 5 = 20

P(C) = |C| / |S|
= 16 / 20
= 4 / 5
= 20 %

$|S| = 25 = 5 + 4 + 16$

$|S| = 4 + 16 = 20$

$|C| = 16$

$$P(C) = \frac{|C|}{|S|} = \frac{16}{20} = \frac{4}{5}$$

$= 20\%$

Q7. Parag Agrawal figures that there is 60% chance Twitter will open a new branch in Hyderabad.
If it happens there is 60% that he will be made CEO again.
What is the probability that Parag Agrawal will become CEO again at Hyderabad.

Event A = Opening a new branch at Hyderabad.
P(A) = 0.6

P(B|A) = Parag Agrawal will be CEO at Hyderabad Branch
P(B|A) = P(A ∩ B) / P(A) = 0.6

P(A) ∩ P(B|A) = 0.6 X 0.6 = 0.36 = 36%

$E(A) =$

$P(A) = 0.6$

$P(B|A) = 0.6$

$\boxed{P(A) \cap P(B|A)}$

$= 0.6 \times 0.6$

$= 0.6 \times 0.6$

$= 0.36 = 36\%$

20

$P(A_1 \cap A_2 A_3 \cap \cdots A_n) =$

Multiplication Theorem: $P(A_1) \cdot P(A_2|A_1) \cdot P(A_3|A_1 \cap A_2) \cdots P(A_n|A_1 \cap A_2 \cdots A$

2

1

P(A ∩ B) = P(A) . P(B|A)

P(A1 ∩ A2 ∩ A3 ∩ … An) = P(A1) . P(A2|A1) . P(A3|A1 ∩ A2) … P(An| A1 ∩ A2 ∩ …An-1)

Q8. A box contains 5 red balls 3 green balls,
You have to pick 2 balls one by one with your eyes closed without replacement.

1) **What is the probability of getting 1st ball is red & 2nd is green?**
2) **What is the probability of getting 3 consecutive red balls?**

Ans1: **Probability of getting 1st ball red and then 2nd ball green**

|S| = 8
Event one (E1) = 1st ball is red
No. of red balls -->|E1| = 5
P(E1) = 5 / 8

After first ball is taken out
|S| = (8 - 1) = 7

Event two (E2|E1) = 2nd ball is green given 1st ball is red
No. of green balls --> 3
P(E2|E1) = 3 / 7

Probability of getting 1st ball red and then 2nd ball green is:
P(E1) . P(E2|E1) = (5/8) X (3/7) = (15/56) = 0.2679 ≈ 27%

Ans2: **Probability of getting 3 consecutive red balls**

|S| = 8
E1 = Red P(E1) = 5/8

|S| = 7
E2 = Red P(E2) = 4/7

|S| = 6
E3 = Red P(E3) = 3/6

Probability of getting 3 consecutive red balls are:
P(E1 E2 E3) = P(E1) X P(E2|E1) X P(E3|E1 ∩ E2)
= (5/8) X (4/7) X (3/6)
= (5/28) = 0.1786 ≈ 18%

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Q9. A fair coin is tossed until head appears for the 1st time,
Probability of number of required tosses is even?
Probability of number of required tosses is odd?

Even:
The no. of tosses is even if the first head appears on the 2nd, 4th, 6th,… toss.

This forms an infinite geometric series

P(Even) = P(X = 2) + P(X = 2) + P(X = 2) + ...
= (1/2)^2 + (1/2)^4 + (1/2)^6 + ...

Formula: Sum of infinite geometric series --> $S = \frac{a}{1 - r}$

Where: a = 1st term of sequence
r = common ratio

Here P(Even) = (1/2)^2 / {1 - (1/2)^2}
= (1/4) / (3/4)
= 1/3

Odd:
Probability of head in 1st attempt: 1/2
Probability of head in 3rd attempt: (1/2) . (1/2) . (1/2) = (1/2)^3
Probability of head in 5th attempt: (1/2) . (1/2) . (1/2) . (1/2) . (1/2) = (1/2)^5

P(Odd) = P(Sample Space) - P(Even)
= 1 - (1/3) = (2/3) = 0.66 = 66%

Q10. There is a lock and bunch of 7 keys,
If we try the keys one by one until the lock opens
What is the probability of lock is opened in 5th attempt?

1st attempt = K1
2nd attempt = K2...
7th attempt = K7

Probability of opening in 1st attempt is --> P(K1) = (1/7)
Probability of not opening in 1st attempt is --> P($\overline{K1}$) = (6/7)

Probability of opening at 4th attempt is --> P($\overline{K1}$. $\overline{K2}$. $\overline{K3}$. K4) = (6/7) . (5/6) . (4/5) . (1/4)
= 1/7
= 14.28 %